

QCalcGeom Simulation Engine v1.0/v1.1: 12-Stage CPT-Symmetric Pipeline with 1.64M eval/s Throughput

Daniel T. Murphy

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PAPER_1152 – QCalcGeom Simulation Engine v1.0/v1.1: 12-Stage CPT-Symmetric Pipeline

Author: Daniel T. Murphy (daniel.murphy00@gmail.com) **Framework:** UQFF v5.77 – Star-Magic Physics **Source:** qcalcgeom_sim_engine.py v1.1.0 (commit e7437feb + 2637b384, May 5 2026) **Integration Date:** May 6, 2026 (Session 234) **Tests:** T_SE_01–T_SE_30; 30/30 PASS **Classification:** QCalcGeom Simulation Engine – Pipeline Architecture

$$\text{throughput} = 1.64 \times 10^6 \text{ eval/s (mean)} \quad 2.01 \times 10^6 \text{ eval/s (peak)}$$

Abstract

This paper registers the QCalcGeom simulation engine (Sessions 202–203) as PAPER_1152, documenting the **12-stage CPT-symmetric numerical pipeline** that transforms raw numeric string batches into fully evaluated UQFF physics observables. The engine processes 21,000 numeric strings $\times$ 23 columns per simulation tick, achieves mean throughput of 1.64×10^6 evaluations per second, and enforces CPT-symmetric forward/backward pair symmetry. Thirty unit tests (T_SE_01–T_SE_30) verify all pipeline stages.

1. Input Format: 23-Column Numeric String

Each simulation tick receives a batch of 21,000 rows. Each row is a whitespace-separated string of 23 numeric columns with indices BCOL_0 through BCOL_22:

Column	Symbol	Physical Quantity
0	t	Simulation time [s]
1	t_n	Normalised time [0, 1]
2	r	Radial distance [m]
3	M	System mass [kg]

Column	Symbol	Physical Quantity
4	B_0	Magnetic field amplitude [T]
5	ω_0	Angular velocity [rad/s]
6	v	Velocity [m/s]
7	ρ_{SCm}	SCm vacuum density [J/m ³]
8	ρ_{UA}	UA vacuum density [J/m ³]
9	E_{vac}	Vacuum energy [J]
10	[SSq]	Self-similar quotient (drift channel)
11	w_{VDS}	VDS branch weight
12	w_{DVP}	DVP branch weight
13	Σ_N	BH26 spectral sum
14	a_{DPM}	DPM acceleration [m/s ²]
15	g_C	Compressed-MUGE gravity [m/s ²]
16	g_R	Resonance-MUGE gravity [m/s ²]
17	g_{BSFG}	BSFG buoyancy gravity [m/s ²]
18	$F_{U_Bi_i}$	Buoyancy-integral force [N]
19	$S_{26}^{(3)}$	Third-order 26D Ramanujan sum
20	G_{phonon}	Phonon linewidth [Γ]
21	w_{oracle}	Wolfram oracle weight
22	[SSq] _{drift}	Accumulated SSq drift

2. 12-Stage Pipeline

Stage 1: READ

Parse raw ASCII numeric strings, validate column count (23), cast to float64.

Stage 2: VDS Branch Evaluation

$$\text{Li}_{26}(z) = \sum_{n=1}^{\infty} \frac{z^n}{n^{26}}, \quad z = [\text{SSq}]$$

$$\text{vds_prime} = \frac{\text{Li}_{25}(z)}{z} \approx 1.0, \quad \text{vds_k_weighted} = \text{Li}_{25}(z) + 25 \cdot \text{Li}_{26}(z)$$

Stage 3: DVP Branch Evaluation

$$\zeta_{\text{DVP}}(z) = \sum_{p>26} \frac{z^{\pi(p)}}{p^{26}}$$

Primes above 26 sieved; dominant term: $a(29) = z^{10}/29^{26}$.

Stage 4: BH26 Spectral Ladder

$$\lambda_k = k(k+25), \quad \Sigma_{N=10} = 1760, \quad E_{\text{Casimir}} = \frac{\hbar f_{\text{RR}}}{2} \sum_{k=1}^N \frac{1}{\lambda_k}$$

Stage 5: DPM Acceleration

$$a_{\text{DPM}} = \frac{F_{\text{DPM}} \cdot f_{\text{DPM}} \cdot E_{\text{vac,neb}}}{c \cdot V_{\text{sys}}}$$

where $F_{\text{DPM}} = I \cdot A \cdot (\omega_1 - \omega_2)$ (di-pseudo-monopole torque).

Stage 6: Compressed-MUGE

$$g_C = a_{\text{DPM}} \cdot \prod_{i=1}^{12} \alpha_i$$

with 12 correction factors: Hubble expansion, magnetic suppression, aether envelope, Ug-sum, cosmological Λ , quantum $\hbar$, Navier-Stokes fluid, perturbation dark matter, and 4 auxiliary resonance terms.

Stage 7: Resonance-MUGE

$$g_R = a_{\text{DPM}} + \sum_{j=1}^{13} a_j^{(\text{res})}$$

where the 13 resonance modes include: THz phonon, vacuum-diffusion, super-frequency, aether-resonance, Ug_{4i}, quantum-frequency, aether-frequency, fluid-frequency, oscillation term, expansion-frequency, f_{TRZ} , and wormhole metric.

Stage 8: BSFG (Buoyancy-Stratified Factorial Geometry)

$$g_{\text{BSFG}} = a_{\text{DPM}} \times \text{joint_coeff} \times \text{vds_k_weighted} \times \frac{\Sigma_{N=10}}{1760}$$

Stage 9: Triple-Point Convergence Validation

Verify $|g_C - g_R|/g_R < 1\%$, $|g_R - g_{\text{BSFG}}|/g_{\text{BSFG}} < 1\%$ simultaneously.

Stage 10: Wolfram Oracle Calibration

Legendre polynomial series $w_k^{(L)}([\text{SSq}])$ for $k = 0, \dots, 8$; cross-calibrated from SOURCE116 hypergraph constants. Provides oracle weight w_{oracle} for Bayesian update of a_{DPM} .

Stage 11: Self-Update

[SSq] drift: +0.000001 per tick (gradient descent at $\kappa = 0.0005$). Learning-rate adaptation via Adam-style schedule.

Stage 12: EXPORT

Write BCOL_22 drift column, serialise tick outputs to JSON/CSV for recall storage.

3. CPT-Symmetric Forward/Backward Pairs

The engine enforces CPT symmetry: for every forward simulation step there exists a backward conjugate:

Property	Forward	Backward
Time	$t > 0$	$t < 0$
t_n	$[0, 1]$	reversed
f_{TRZ}	+	−
$\cos(\pi t_n)$	$+1 \rightarrow -1$	$-1 \rightarrow +1$

The closure condition: for forward phase ($D_A = +3$) and backward phase ($D_B = -3$), net displacement $D_A + D_B = 0$ (proven closed loop, see PAPER_1153).

4. Scale Separation Results

At [SSq] = 0.57, standard tick (t=1 day, r=1 AU, M= $M_\odot$):

Pair	Orders of Magnitude
Compressed-MUGE vs Wolfram oracle	35.85 OOM
Resonance-MUGE vs Wolfram oracle	94.17 OOM
Resonance-MUGE vs BSFG	6.79 OOM

The 6.79 OOM Resonance-BSFG separation is the key calibration target: it corresponds to the ratio $g_R/g_{\text{BSFG}} \approx 6.17 \times 10^6$ and sets the regime boundary between quasi-periodic and coherent-buoyancy gravity modes.

5. Throughput Benchmark

Metric	Value
Batch size	21,000 strings $\times$ 23 cols
Mean throughput	1.64×10^6 eval/s
Peak throughput	2.01×10^6 eval/s
Target (design)	1×10^6 eval/s
Achieved / Target	$1.64 \times$ nominal, $2.01 \times$ peak

The LUT (look-up table) precomputation in `_Precompute` reduces VDS/DVP evaluations to array indexing, providing the $1.64 \times$ performance margin.

6. Test Suite T_SE_01–T_SE_30

Test	Stage	Assertion
T_SE_01–05	READ	Parse 23 cols; reject malformed
T_SE_06–10	VDS	$\text{vds_prime} \in (0.9, 1.1)$
T_SE_11–15	DVP/BH26	$\Sigma_{10} = 1760; a(29) > 0$
T_SE_16–20	DPM+MUGE	$g_C, g_R, g_{\text{BSFG}} > 0$
T_SE_21–25	CPT	$ D_A + D_B < \epsilon$
T_SE_26–30	Throughput	$\geq 1 \times 10^6 \text{ eval/s}$

Result: 30/30 tests PASS

7. Conclusions

The QCalcGeom simulation engine provides the first fully CPT-symmetric, 12-stage pipeline for simultaneous evaluation of VDS, DVP, BH26, MUGE Compressed, MUGE Resonance, and BSFG gravity branches. Mean throughput $1.64\times$ the design target confirms production readiness.

References

1. Murphy, D.T. (2026). *qcalcgeom_sim_engine.py v1.1.0*. Star-Magic repository, commits e7437feb, 2637b384.
2. Murphy, D.T. (2026). *PAPER_1151: VDS/DVP/BH26 Variant Branches*. Session 202.
3. Murphy, D.T. (2025). *PAPER_429: BSH Buoyancy Harmonic Series*.
4. Murphy, D.T. (2026). *Star-Magic.txt Chapter 4: DPM and MUGE Framework*.
5. Murphy, D.T. (2026). *PAPER_1153: Primordial Timing Function Net=0 Proof*. Session 203.

*Updated: 2026-05-06 (Session 234, PAPER_1152). Compliant: CVW v2.0.0, G6 SM Anchor Gate.
Author: Daniel T. Murphy*